

How a document travels from scan to filed

Every scan runs down the same conveyor belt. Each station either fills in a field, checks one, or decides where the document goes next. This shows what each station does, a real example, and why it exists.

We follow one running example the whole way down: a **scanned invoice from “City Office NI”, invoice No. 152574, dated 15-12-2025, total £101.28.**

The one idea to hold onto: Scan Finder never trusts a single guess blindly. It reads each field several different ways, prefers the most trustworthy source, and — whenever two good methods *disagree* — it hands the document to a person instead of filing a mistake.

It also learns. Every correction a person makes teaches it where a sender’s fields sit and how their names are spelled, so the same sender reads better next time.

- Bringing it in
- Reading the page
- Finding the fields
- Checking the values
- A decision point
- A person’s turn
- Filed away

STAGE BY STAGE — TOP TO BOTTOM

1 The document arrives

BRINGING IT IN

You either import a file yourself, or drop it into a folder Scan Finder is watching. It immediately takes its own working copy so your original is never altered or moved by mistake.

EXAMPLE

Someone drops `invoice_scan.pdf` into the watched folder. Scan Finder copies it into its own inbox and starts work on the copy.

WHY

Working on a safe copy means nothing you own is ever at risk, and the app can re-do its work later without needing the original again.

2 Straighten the page Auto-rotate

READING THE PAGE

If the scan went in sideways or upside-down, it’s quietly turned the right way up first. It only does this when it’s confident — it never guesses.

EXAMPLE

A page fed in rotated 90° is rotated back to upright before anything tries to read it.

WHY

Text recognition is far more accurate on an upright page. The untouched original is still kept, so a rare wrong turn is always recoverable.

3 Read the words off the page OCR

READING THE PAGE

This turns the picture into text — and, importantly, records *where* each word sits on the page. There's a fork here:

PDF MADE BY SOFTWARE

Read the exact text straight out of the file — perfect, no guessing.

A PHOTO / SCAN

Use OCR (Tesseract) to recognise the letters from the image.

EXAMPLE

The picture of **Invoice No. 152574** becomes the text "Invoice No. 152574", and the app notes exactly where on the page those words are.

WHY

Everything after this needs words, not pixels. A software-made PDF is read perfectly; a scan needs OCR. Knowing each word's position is what lets later stages read "the value next to this label."



4 Have I seen this layout before? Stage 0 · template match

FINDING THE FIELDS

It compares the sender's logo and a fingerprint of the page's wording against layouts it has learned. A match means it already knows roughly where this sender puts each field.

EXAMPLE

The logo matches "City Office NI", so it starts with a head-start: it knows their invoice number usually sits top-right.

WHY

Recognising a familiar sender lets it go straight to the right spots instead of hunting from scratch — this is a big part of "getting smarter over time."



5 Read the spots a person taught it Stage 0.5 · taught zones

FINDING THE FIELDS

If someone has drawn "the invoice number lives here" boxes for this layout, it reads those exact zones. If the scan is shifted a little, it re-aligns the page first, then reads — "line it up, then read."

EXAMPLE

An admin once drew a box around City Office's invoice number. Even on a slightly shifted new scan, the box is nudged to line up and read there.

WHY

A human-taught location is the most trustworthy signal there is, so it's tried early and given a lot of weight.



6 Look next to the labels Stage 1 · keywords

FINDING THE FIELDS

It scans for ordinary labels — "Invoice No", "Date", "Total" — and reads the value sitting next to each. It's careful about look-alikes (it won't read "Subtotal" when it wants "Total").

EXAMPLE

Sees **Total £101.28** and records the total as 101.28 — and knows the £-figure beside "Subtotal" is a different field.

WHY

Most documents label their own fields, so this alone fills roughly two-thirds of fields with no teaching needed at all.



7

Use what I've learned about this sender

Stage 2 · learned
positions

FINDING THE FIELDS

improved

It applies positions learned from past corrections, and recovers gracefully when a page has drifted. This stage also holds a new safety check for taught reference/invoice numbers:

Does the tightly-cropped read of a taught number **agree** with reading the same label off the whole page?

THEY AGREE

Trust it and carry on — nothing changes.

THEY DISAGREE

Prefer the whole-page value and send it to a person. Never file a silent mis-read.

EXAMPLE

A tight crop misreads a digit as **192074**, but the full page plainly says **152574**. Rather than file 192074 at 97% confidence, it uses 152574 and flags it to check.

WHY

A zoomed-in crop can occasionally mangle one digit. Cross-checking it against the full page stops a confident-but-wrong number from filing itself unnoticed.

8

Fix known mis-reads

Stage 2.5 · learned corrections

FINDING THE FIELDS

If it has learned that a particular sender's scans have a habit (say, a zero that always reads as the letter "O"), it applies that specific fix here.

EXAMPLE

A sender whose account codes read **S02** instead of **S0Z** get quietly put right, because that mistake has been corrected before.

WHY

Some mistakes are predictable per sender. Learning them means the same one doesn't need correcting twice.

A quiet aside — the "AI reader" (Stage 3). There's an optional large-language-model reader that can be called in for the hardest pages. In the shipped app it stays switched off and isn't part of normal filing — everything above is rule-based and runs entirely on your PC. It's mentioned only so the picture is complete.

9

Sanity-check the values & the maths

Stage 4 · validation

CHECKING THE VALUES

Dates are tidied into one consistent format, currencies are worked out, and the numbers are checked against each other — do the line items add up to the total?

EXAMPLE

"Aug 03 2012" becomes **03-08-2012**. If a stated total doesn't balance against the lines, it reconsiders it — e.g. correcting a drifted total to the figure that actually adds up (this fixed our £101.28 example).

WHY

Catching impossible values and settling on one date format keeps the filename, the stored value, and future learning all in agreement.

10 Does it look like this sender usually looks?

Stage 4.5 · consistency & confidence

CHECKING THE VALUES

Each value is compared to this sender's own history. A garbled but recognisable company name is repaired to its known spelling; anything that looks off is flagged rather than silently accepted. Finally, an overall "how sure am I?" score is worked out.

EXAMPLE

A supplier read as PROFLE is repaired to "Profile" from the learned spelling; a name that comes out suspiciously short gets a gentle "please verify."

WHY

The sender's own past documents are the best yardstick. A flagged value a person can glance at is always better than a confident wrong one that slips through.

? Confident enough to file on its own?

Are all the required fields filled, nothing flagged for a person, and the overall score at or above your chosen threshold?

YES → FILE IT AUTOMATICALLY

The document skips the queue and files itself. You can set how strict this is (default: only perfectly-clean documents auto-file).

NO → SEND TO REVIEW

Anything missing, flagged, or below the bar waits for a person. Nothing uncertain is ever filed unseen.

11 A person checks it Review

A PERSON'S TURN

In the Review screen someone confirms the fields or corrects anything wrong. Those corrections are the fuel for learning — where fields sit, how names are spelled, which value was right.

EXAMPLE

The reviewer confirms 152574 and tidies the supplier name. Next time City Office sends an invoice, both read correctly on their own.

WHY

This is the moment the system gets smarter. A minute spent correcting today removes the same correction on every future document from that sender.

12 File it away, tidily Filing

FILED AWAY

The document is moved into a clean folder structure with a consistent, searchable name, alongside a small hidden data file that remembers what each field was.

EXAMPLE

City Office NI / 2025 / December / Invoice.15-12-2025.152574.pdf

WHY

Consistent names and folders are the whole point — every filed document is instantly findable later, by anyone.

🔄 The learning loop — why it improves

Every correction a person makes in **Review** feeds straight back into the “finding the fields” stages (4, 5, 6, 7, 8): the taught position, the corrected spelling, and the confirmed value all become part of what that sender “usually looks like.” So the more you use it, the more documents sail through and file themselves — and the fewer ever reach the Review queue.

Scan Finder — internal plain-English overview of the detection pipeline. One running example used throughout (City Office NI invoice 152574). The technical names in small grey text (“Stage 2”, “OCR”, etc.) map to the engine’s real stages if you ever want to go deeper.